

TrES-3b

R-2828

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130814 · created 2026-09-05T23:25:06+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456518.7607 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.147 +/- 0.055
Transit depth (Rp/Rs)^2	2.16 +/- 1.63 [%]
Orbital Inclination (inc)	86.96 +/- 2.78
Airmass coefficient 1 (a1)	0.758 +/- 0.039
Airmass coefficient 2 (a2)	0.207 +/- 0.069
Scatter in the residuals of the lightcurve fit is	5.24 %
Best Comparison Star	#1 - [476, 426]
Optimal Aperture	3.39
Optimal Annulus	9.28
Transit Duration (day)	0.077 +/- 0.0099

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.147 $\pm$ 0.055 vs 0.16309 (deviation 0.28 σ)
Duration fit vs published	0.077 d vs 0.0567 d (deviation 1.99 σ)
Mid-time O–C	0.5 min
Depth SNR	2.6
Residual scatter	5.24 %

Light curve

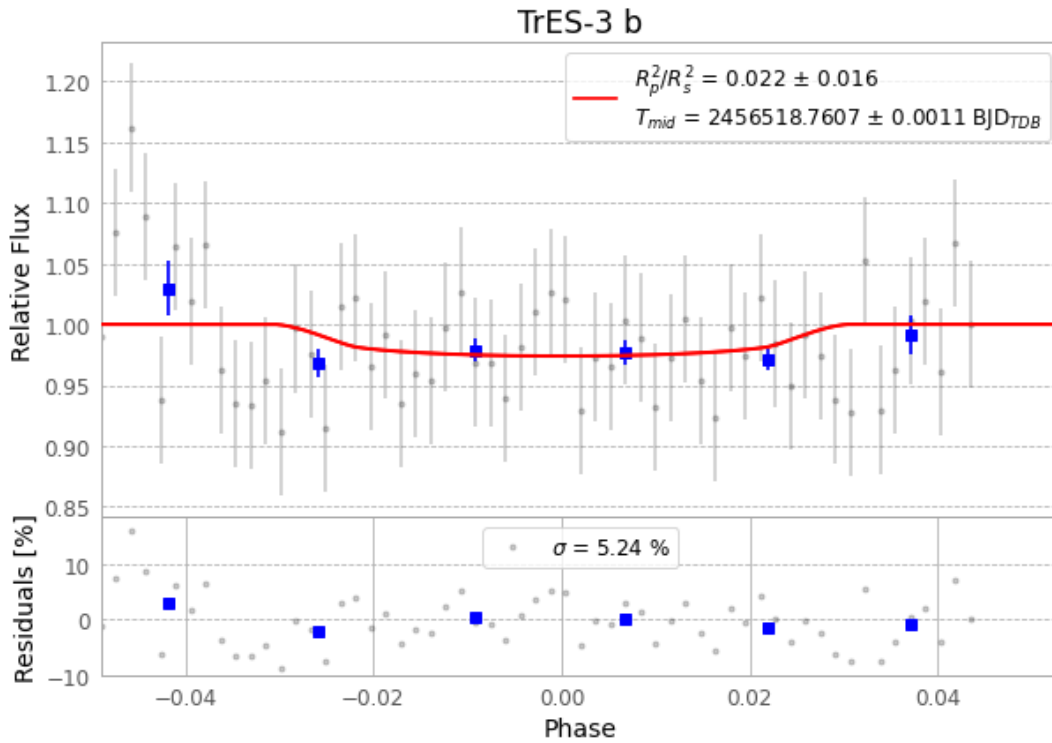


Figure 1 — FinalLightCurve_TrES-3 b_2013-08-13.png (SHA-256 eb66abf1a293574b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [368, 272], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 00b5efcbb3754f48...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), TRES-3130814043912.FITS → TRES-3130814075113.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130814.log (ff68875d145a...), FinalLightCurve_TrES-3 b_2013-08-13.png (eb66abf1a293...), FinalLightCurve_TrES-3 b_2013-08-13.csv (5315b4ddb1b3...)

Compute resources

Wall time 515.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 23:25Z.